Table 1. *BAP1* missense variants described in this study.

All 58 missense variants shown in Figure 1a: the UDN proband variant (p.Asn229Lys), the 17 KURIS/NDD pathogenic controls, the ClinVar pathogenic/likely-pathogenic missense variants associated with *BAP1* tumor predisposition syndrome (TPDS), and the 28 ClinVar benign/likely-benign missense controls. The single-letter notation is the variant label used in Figures 1 and 2. SGE functional scores, ClinVar classifications, and the clinical evidence for each variant are in Supplementary Tables 1 and 2.

Gene; HGNC ID	Category	HGVS g.	HGVS c.	HGVS p.	Variant
BAP1; HGNC:950	KURIS/NDD (proband)	NC_000003.12:g.52406349G>T	NM_004656.4:c.687C>A	NP_004647.1:p.Asn229Lys	N229K
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52409851T>G	NM_004656.4:c.28A>C	NP_004647.1:p.Ser10Arg	S10R
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52409851T>C	NM_004656.4:c.28A>G	NP_004647.1:p.Ser10Gly	S10G
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52409845G>T	NM_004656.4:c.34C>A	NP_004647.1:p.Pro12Thr	P12T
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52409845G>C	NM_004656.4:c.34C>G	NP_004647.1:p.Pro12Ala	P12A
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52409585C>T	NM_004656.4:c.91G>A	NP_004647.1:p.Glu31Lys	E31K
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52409578T>G	NM_004656.4:c.98A>C	NP_004647.1:p.Tyr33Ser	Y33S
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52408583A>G	NM_004656.4:c.146T>C	NP_004647.1:p.Leu49Pro	L49P
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52408553C>G	NM_004656.4:c.176G>C	NP_004647.1:p.Arg59Pro	R59P
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52408553C>T	NM_004656.4:c.176G>A	NP_004647.1:p.Arg59Gln	R59Q
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52408550C>T	NM_004656.4:c.179G>A	NP_004647.1:p.Arg60Gln	R60Q
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52408061C>G	NM_004656.4:c.272G>C	NP_004647.1:p.Cys91Ser	C91S
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52408062A>C	NM_004656.4:c.271T>G	NP_004647.1:p.Cys91Gly	C91G
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52408062A>G	NM_004656.4:c.271T>C	NP_004647.1:p.Cys91Arg	C91R
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52408058G>A	NM_004656.4:c.275C>T	NP_004647.1:p.Ala92Val	A92V
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52407247G>C	NM_004656.4:c.507C>G	NP_004647.1:p.His169Gln	H169Q
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52407248T>C	NM_004656.4:c.506A>G	NP_004647.1:p.His169Arg	H169R
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52402325C>T	NM_004656.4:c.2153G>A	NP_004647.1:p.Arg718Gln	R718Q
BAP1; HGNC:950	Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52409848C>G	NM_004656.4:c.31G>C	NP_004647.1:p.Asp11His	D11H
BAP1; HGNC:950	Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52408596C>T	NM_004656.4:c.133G>A	NP_004647.1:p.Gly45Arg	G45R
BAP1; HGNC:950	Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52407958C>G	NM_004656.4:c.375G>C	NP_004647.1:p.Glu125Asp	E125D
BAP1; HGNC:950	Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52407454C>T	NM_004656.4:c.382G>A	NP_004647.1:p.Gly128Arg	G128R
BAP1; HGNC:950	Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52407415G>A	NM_004656.4:c.421C>T	NP_004647.1:p.His141Tyr	H141Y
BAP1; HGNC:950	Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52407399C>G	NM_004656.4:c.437G>C	NP_004647.1:p.Arg146Thr	R146T
BAP1; HGNC:950	Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52407215A>G	NM_004656.4:c.539T>C	NP_004647.1:p.Leu180Pro	L180P
BAP1; HGNC:950	Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52406902A>G	NM_004656.4:c.586T>C	NP_004647.1:p.Trp196Arg	W196R
BAP1; HGNC:950	Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52406884A>G	NM_004656.4:c.604T>C	NP_004647.1:p.Trp202Arg	W202R
BAP1; HGNC:950	Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52406882C>G	NM_004656.4:c.606G>C	NP_004647.1:p.Trp202Cys	W202C
BAP1; HGNC:950	Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52403437G>C	NM_004656.4:c.1708C>G	NP_004647.1:p.Leu570Val	L570V
BAP1; HGNC:950	Cancer predisposition (ClinVar P/LP)	NC_000003.12:g.52402643T>C	NM_004656.4:c.2015A>G	NP_004647.1:p.Asp672Gly	D672G
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52409574G>C	NM_004656.4:c.102C>G	NP_004647.1:p.Asp34Glu	D34E
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52409555C>T	NM_004656.4:c.121G>A	NP_004647.1:p.Gly41Ser	G41S
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52408590T>C	NM_004656.4:c.139A>G	NP_004647.1:p.Ile47Val	I47V
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52408493T>C	NM_004656.4:c.236A>G	NP_004647.1:p.Asn79Ser	N79S

Gene; HGNC ID	Category	HGVS g.	HGVS c.	HGVS p.	Variant
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52407990T>G	NM_004656.4:c.343A>C	NP_004647.1:p.Met115Leu	M115L
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52407418C>T	NM_004656.4:c.418G>A	NP_004647.1:p.Ala140Thr	A140T
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52405827T>G	NM_004656.4:c.869A>C	NP_004647.1:p.Asn290Thr	N290T
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52405827T>C	NM_004656.4:c.869A>G	NP_004647.1:p.Asn290Ser	N290S
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52405812A>G	NM_004656.4:c.884T>C	NP_004647.1:p.Val295Ala	V295A
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52405795C>A	NM_004656.4:c.901G>T	NP_004647.1:p.Ala301Ser	A301S
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52405791G>A	NM_004656.4:c.905C>T	NP_004647.1:p.Pro302Leu	P302L
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52405279G>T	NM_004656.4:c.947C>A	NP_004647.1:p.Ala316Glu	A316E
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52404549C>G	NM_004656.4:c.1154G>C	NP_004647.1:p.Arg385Pro	R385P
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52404520A>C	NM_004656.4:c.1183T>G	NP_004647.1:p.Ser395Ala	S395A
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52404485C>G	NM_004656.4:c.1218G>C	NP_004647.1:p.Glu406Asp	E406D
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52403877G>T	NM_004656.4:c.1268C>A	NP_004647.1:p.Thr423Lys	T423K
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52403799G>A	NM_004656.4:c.1346C>T	NP_004647.1:p.Ala449Val	A449V
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52403718A>G	NM_004656.4:c.1427T>C	NP_004647.1:p.Val476Ala	V476A
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52403646C>T	NM_004656.4:c.1499G>A	NP_004647.1:p.Gly500Asp	G500D
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52403292C>T	NM_004656.4:c.1736G>A	NP_004647.1:p.Gly579Glu	G579E
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52403274G>A	NM_004656.4:c.1754C>T	NP_004647.1:p.Ser585Phe	S585F
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52403242T>C	NM_004656.4:c.1786A>G	NP_004647.1:p.Ser596Gly	S596G
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52403235G>A	NM_004656.4:c.1793C>T	NP_004647.1:p.Pro598Leu	P598L
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52403199C>T	NM_004656.4:c.1829G>A	NP_004647.1:p.Arg610Lys	R610K
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52403190G>A	NM_004656.4:c.1838C>T	NP_004647.1:p.Thr613Met	T613M
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52403162C>G	NM_004656.4:c.1866G>C	NP_004647.1:p.Leu622Phe	L622F
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52402827G>C	NM_004656.4:c.1935C>G	NP_004647.1:p.Asn645Lys	N645K
BAP1; HGNC:950	ClinVar B/LB (control)	NC_000003.12:g.52402828T>C	NM_004656.4:c.1934A>G	NP_004647.1:p.Asn645Ser	N645S

HGVS descriptions use the reference sequences NC_000003.12 (GRCh38), NM_004656.4 (RefSeq MANE Select transcript), and NP_004647.1 (protein) for BAP1 (HGNC:950).